

EN3150 Assignment 01
Learning from data and related challenges and
linear models for regression
B.Sc. Engineering, Semester 05

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1 Data Pre-processing and Dealing with Skewness

1. Use the code given in Listing 1 to generate two distinct features. Use your index no as an input.

```
import numpy as np
import matplotlib.pyplot as plt

def generate_features(n_samples=1000, student_id
                    =12345):
    np.random.seed(student_id % 1000)

    # Feature 1: Normally distributed data
    feature_1 = np.random.normal(loc=50, scale=15, size=
                                n_samples)

    # Feature 2: Exponentially distributed data (highly
    # skewed)
    feature_2 = np.random.exponential(scale=2.0, size=
                                n_samples)

    # Introduce a few extreme artificial outliers to
    # Feature 2
    feature_2[np.random.choice(n_samples, 5, replace=False
                                )] = np.random.uniform(50, 100, 5)

    return feature_1, feature_2

student_id = 40266 # Replace with your index number
f1, f2 = generate_features(student_id=student_id)
```

```
plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.hist(f1, bins=50, color='blue', alpha=0.7)
plt.title("Feature 1 Distribution")
plt.subplot(1, 2, 2)
plt.hist(f2, bins=50, color='red', alpha=0.7)
plt.title("Feature 2 Distribution")
plt.show()
```

Listing 1: Feature generation.

2. Plot the histograms of both features.
3. Apply standard scaling (Z-score normalization) to both features. Plot the resulting histograms.
4. Observe the scaled distribution of Feature 2. Does standard scaling effectively normalize highly skewed data with extreme outliers? Justify your answer using the mathematical definition of standard scaling. [10 marks]
5. Propose and apply a non-linear mathematical transformation (e.g., logarithmic transformation) to Feature 2 *before* applying standard scaling. Plot the new distribution and discuss why this two-step approach is more effective for exponentially distributed data. [15 marks]

2 Linear Regression and Regularization

Consider a dataset predicting housing prices (y) based on several highly correlated features ($\mathbf{x}$). You train a standard Multiple Linear Regression model and compute the Residual Sum of Squares (RSS).

To prevent overfitting, you decide to apply Ridge Regression (L_2 Regularization), which modifies the standard loss function to:

$$L(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N (y_i - \mathbf{w}^T \mathbf{x}_i)^2 + \lambda \sum_{j=1}^d w_j^2 \quad (1)$$

Where N is the number of samples, d is the number of features, $\mathbf{w}$ contains the model weights, and $\lambda \geq 0$ is the regularization hyper-parameter.

1. Load the diabetes dataset from `sklearn.datasets` using `load_diabetes()`. Use all features as independent variables and the target progression metric as the dependent variable. Split the data into 80% training and 20% testing sets. [5 marks]
2. Train a standard Linear Regression model ($\lambda = 0$). Calculate and report the following metrics on the **testing set**:

- Residual Standard Error (RSE)
- Mean Squared Error (MSE)
- R^2 statistic

[10 marks]

3. Train three separate Ridge Regression models using `sklearn.linear_model.Ridge` with $\lambda \in \{0.1, 1.0, 10.0\}$.
4. Plot the magnitude of the estimated coefficients ($\mathbf{w}$) for all four models (Standard LR, Ridge $\lambda = 0.1, 1.0, 10.0$) on a single graph. [15 marks]
5. What happens to the coefficient values as $\lambda \rightarrow \infty$? Based on this, explain how Ridge Regression acts as a robust estimator against highly correlated features. [15 marks]

3 Robust Regression and the Huber Loss

You are given a dataset containing a severe outlier, shown in Table 1.

Table 1: 2D Dataset with an Outlier

i	x_i	y_i
1	1.0	2.5
2	2.0	4.1
3	3.0	6.2
4	4.0	8.0
5	5.0	35.0 (Outlier)
6	6.0	12.1

You are evaluating two candidate linear models:

- **Model A (Standard OLS fit):** $y = 3.8x - 2.0$
- **Model B (Trend-fitting fit):** $y = 2.0x + 0.2$

Instead of standard Mean Squared Error (MSE), you will evaluate these models using the **Huber Loss**, which is designed to be robust to outliers. The Huber loss for a single data point is defined as:

$$L_{\delta}(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & \text{for } |y - \hat{y}| \leq \delta, \\ \delta|y - \hat{y}| - \frac{1}{2}\delta^2 & \text{for } |y - \hat{y}| > \delta. \end{cases} \quad (2)$$

Where δ is a hyper-parameter.

1. Calculate the standard Mean Squared Error (MSE) for both Model A and Model B using the data in Table 1. Which model has the lower MSE? [10 marks]

2. Calculate the average Huber Loss for both Model A and Model B using $\delta = 3.0$. Which model performs better under the Huber Loss? [20 marks]
3. Explain the mechanism by which the Huber loss mathematically limits the influence of the outlier at $x_5 = 5.0$ compared to the standard MSE. Use your calculated values to justify your answer. [15 marks]

4 Loss Functions Comparison

You are tasked with designing the loss function architecture for a machine learning model that predicts the probability of a patient having a specific disease based on medical imagery. The output of your model $\hat{y}$ is a probability value bounded between 0 and 1. The ground truth y is binary, where $y \in \{0, 1\}$.

The Binary Cross Entropy (BCE) loss is defined as:

$$\text{BCE} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (3)$$

1. Briefly explain the role of the logarithmic terms in the BCE formula when the true label y_i is 1 versus when it is 0. [5 marks]
2. A junior engineer suggests using Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (4)$$

instead of BCE to train this probability model, arguing it is computationally simpler. Calculate the loss for a single positive sample ($y = 1$) when the model predicts $\hat{y} = 0.01$ using both MAE and BCE. [10 marks]

3. Based on your calculations, explain why BCE is superior to MAE for classification tasks involving probabilities where true value is either 1 or 0. [15 marks]

5 Submission

- Upload a report as a pdf file named as "your_indexno_EN3150_A01.pdf". Include the index number and the name within the report as well. Please include all your answers in the report.
- Pay careful attention to formatting such as font size, spacing, and margins.
- Include a title page with necessary information (e.g., title, author, date, index no).
- Use consistent and professional formatting throughout the document.

- Plagiarism will be checked and in cases of plagiarism, an extra penalty of 50% will be applied. In case of copying from each other, both parties involved will receive a grade of zero for the assignment. Academic integrity is of utmost importance, and any form of plagiarism¹ or cheating will not be tolerated.
- An extra penalty of 15% is applied for late submission.

¹<https://en.wikipedia.org/wiki/Plagiarism>